Unit III: Cell biology and Biomolecules

Chapter 8

Biomolecules

Learning Objectives

The learner will be able to,

- *Recognise the basic structure of carbohydrates, proteins, lipids and nucleic acids and differentiate the various pattern of classification with respect to structure.*
- *Familiarise with the general structure of amino acids and its classification based on the functional group.*
- *know the structure and classification of enzymes properties*
- *Understand lipids as a biomolecule and discuss the properties of lipids.*
- *Have a deeper knowledge about structure of nucleic acids.*

Chapter Outline

8.1 Water
8.2 Primary and Secondary Metabolites
8.3 Carbohydrates
8.4 Lipids
8.5 Proteins
8.6 Enzymes
8.7 Nucleic Acids

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Having learnt the structure of the cell, we can now understand that each component of the cell is responsible for a specific function. The cell components are made of collection of molecules called as **cellular pool**, which consists of both inorganic and organic compounds. Inorganic compounds include salts, mineral ions and water.

Organic compounds include carbohydrates, lipids, amino acids, proteins, nucleotides, hormones and vitamins. Some organic molecules remain in colloidal form in the aqueous intracellular fluid. Others exist in non-aqueous phases like the lipid membrane and cell walls. The cell maintains this pool by the intake and elimination of specific molecules (Figure 8.1).

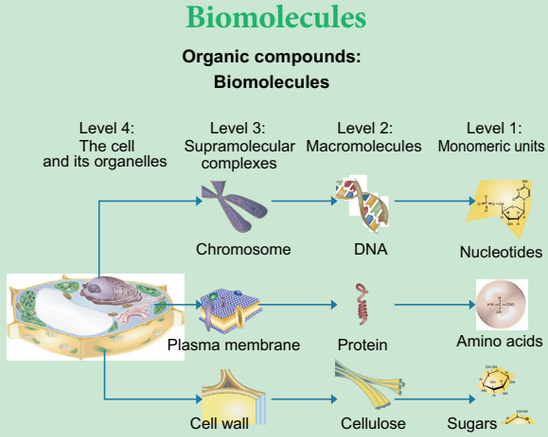


Figure 8.1: Components of cell

The minerals essential for plant growth are of two types: **macronutrients**, which are required in larger amounts (Eg. potassium, phosphorus, calcium, magnesium, sulphur and iron) and **micronutrients**, which are required in trace amounts Eg. cobalt, zinc, boron, copper, molybdenum and manganese are essential for enzyme action. Example,

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